

Graphical Abstract

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Highlights

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- Propose an LLM-empowered semantic generation module to alleviate the semantic gap problem in traditional ID-based recommendation models.
- Granular ball computing is introduced into the interest modeling process of recommendation systems to enhance the model's robustness in sparse and noisy environments.
- Design a dynamic mask-based graph denoising learning mechanism to enhance the stability and reliability of representation learning.
- Experiments on three datasets demonstrate the superior performance of the proposed model.

LLM-GRec: Large Language Model Enhanced Graph Recommendation via Granular-Ball Computing

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ABSTRACT

Large Language Models (LLMs) have demonstrated strong capabilities in semantic understanding and have been widely introduced into recommendation systems to alleviate the issue of insufficient semantic information in traditional collaborative filtering. However, existing methods still face two key challenges in interest modeling. First, they lack effective utilization of collaborative signals, resulting in incomplete interest representations. Second, the interests generated by LLMs are typically represented as discrete phrases, which are susceptible to noise and lack stable structural organization. Moreover, semantic noise may be progressively amplified during graph propagation, thereby degrading recommendation performance. To address these issues, we propose a robust graph-based recommendation framework, namely LLM-GRec, which enhances the reliability of interest modeling while fully leveraging the semantic capabilities of LLMs. Specifically, we design a collaborative-enhanced interest generation mechanism that integrates textual information with collaborative signals to obtain more comprehensive semantic representations. Furthermore, we introduce granular-ball computing for multi-granularity interest modeling to learn robust structural representations of user interests. In addition, a dynamic mask-driven denoising strategy is proposed to effectively suppress the propagation of semantic noise. Extensive experiments on three real-world datasets demonstrate that LLM-GRec consistently outperforms state-of-the-art methods across various metrics, achieving an average improvement of 4.3% over the strongest baseline on the best-performing dataset, and showing more significant advantages in sparse scenarios. Our source code is available at <https://github.com/shenjiale534/LLM-GRec>

1. Introduction

Personalized recommender systems have become a key technique for alleviating information overload [12]. In recent years, graph neural networks (GNNs) have emerged as an important paradigm in recommender systems due to their superior capability in modeling non-Euclidean data structures [23]. By recursively aggregating neighborhood information on the user-item bipartite graph, advanced models such as NGCF [17] and LightGCN [5] are able to effectively capture high-order collaborative signals, thereby achieving significant improvements in recommendation accuracy. Despite their effectiveness, GNN-based methods still face several non-negligible challenges in real-world applications. A primary issue is the semantic gap caused by data sparsity. Most existing GNN-based recommender models rely on anonymous ID embeddings to represent users and items, lacking an understanding of the underlying semantic intentions behind user behaviors. This limitation becomes particularly severe in sparse interaction scenarios [8]. To address this issue, recent studies have attempted to incorporate LLMs to enhance the semantic representation capability of recommender systems [10, 13]. Benefiting from their strong knowledge reasoning and semantic understanding abilities, LLMs can extract richer interest information from user historical behaviors, thereby compensating for the limitations

of traditional collaborative filtering methods in semantic modeling.

However, existing LLM-based recommendation methods still face several critical challenges in interest modeling. First, most approaches primarily rely on textual descriptions derived from user historical behaviors to generate interest representations, while lacking effective utilization of collaborative signals. Since user interaction data is typically sparse, relying solely on textual information may lead to incomplete interest modeling, thereby degrading recommendation performance. Second, the interests generated by LLMs are usually represented as discrete phrases or keywords. Such representations are prone to noise and redundancy and lack stable structural organization. Existing methods often adopt fine-grained point-wise representations or conventional clustering techniques, which are sensitive to noisy or anomalous data under sparse or noisy conditions, making it difficult to learn robust interest structure representations. In addition, LLMs may suffer from the hallucination phenomenon when generating semantic content, producing interest signals that are inconsistent with users' true preferences [7]. Once such noisy semantics are introduced into the graph structure and involved in multi-hop propagation, they may be continuously amplified, thereby negatively affecting representation learning. Existing graph-based recommendation methods typically rely on fine-grained interactions for information propagation and are highly sensitive to noise [22], which further exacerbates this issue.

To address the above challenges, this paper proposes a novel recommendation framework, termed **LLM-GRec**,

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which aims to effectively integrate the semantic reasoning capability of LLMs with collaborative signals from graph structures, while mitigating the negative impact of semantic noise. Specifically, our approach consists of three tightly coupled components. (1) First, we design an LLM-enhanced semantic generation module that transforms sparse user interactions into explicit and semantically rich interest descriptions by leveraging the knowledge capacity of LLMs, thereby constructing an augmented semantic–collaborative graph. (2) Second, to extract robust features from complex textual and structural signals, we introduce granular-ball computing into the recommendation framework. Instead of conventional fine-grained point embeddings, we employ adaptive hyperspheres to perform multi-granularity clustering over user interests. This coarse-grained modeling strategy enables better coverage of the data space and improves robustness under sparse conditions. (3) Finally, to address the semantic noise introduced by LLMs, we propose a dynamic mask-based graph denoising learning mechanism. By adaptively regulating graph propagation paths, the proposed method effectively suppresses the propagation of spurious interests and enhances the reliability of learned representations.

In summary, the main contributions of this work are as follows:

- We propose an LLM-enhanced semantic generation module that transforms implicit collaborative interaction signals into explicit semantic interests, effectively alleviating the semantic gap in traditional ID-based recommendation models.
- We introduce granular-ball computing into interest modeling for recommender systems. By performing multi-granularity clustering over user interests, the proposed method learns robust interest structure representations and enhances model robustness under sparse and noisy environments.
- We design a dynamic mask-based graph denoising learning mechanism that effectively suppresses noise propagation introduced by LLM-generated interests, thereby improving the stability and reliability of representation learning.
- Extensive experiments on three real-world datasets demonstrate that LLM-GRec consistently outperforms state-of-the-art methods across multiple evaluation metrics, with particularly significant improvements under sparse scenarios.

2. Related Work

2.1. LLM-based Recommendation Methods

In recent years, LLMs have achieved remarkable progress in natural language understanding and knowledge reasoning, making them an emerging paradigm in recommender system research. Traditional recommender systems typically rely on user–item interaction data for collaborative filtering, where

ID-based representations often fail to capture the underlying semantic information behind user behaviors. To address this limitation, an increasing number of studies have explored leveraging the semantic understanding capability of LLMs to enhance representation learning in recommender systems.

Early approaches primarily utilize LLMs to extract semantic features from textual content and integrate them with traditional collaborative filtering signals. RLMRec [13] employs LLMs to derive semantic representations of users and items, which are jointly learned with collaborative filtering features to construct enriched user interest representations. LLMRec [10] further utilizes LLMs to generate semantic-enhanced information and augment the user–item graph structure, thereby improving the representation capability of graph-based recommendation models. TALLRec [30] designs a parameter-efficient fine-tuning framework tailored for recommendation tasks, enabling LLMs to better adapt to recommendation scenarios and enhance overall performance.

Despite these advances in semantic modeling, LLMs may suffer from the hallucination problem, where generated semantic information is inconsistent with users’ true preferences [7]. Such noisy information can negatively affect representation learning in recommender systems. To mitigate this issue, LLM-ESR [11] enhances robustness in complex scenarios by leveraging LLMs to improve semantic representations in long-tail sequence recommendation, thereby improving performance under sparse data conditions. However, these methods still predominantly rely on fine-grained semantic representations and lack explicit modeling and structural constraints for noisy semantics. As a result, they struggle to fundamentally suppress the propagation and amplification of erroneous interest signals introduced by LLMs in graph structures. Therefore, how to effectively leverage LLMs while filtering noise and learning robust interest structure representations remains an open problem.

2.2. Multi-granularity Interest Modeling

In real-world recommendation scenarios, user interests are inherently diverse, and representing them with a single vector is often insufficient to capture complex user preferences. As a result, multi-interest recommendation has become an important research direction. Early studies mainly focus on learning multiple interest representations to model diverse user preferences. MIND [9] extracts multiple interest vectors from user historical behaviors through a dynamic routing mechanism, enabling the modeling of multi-interest structures. ComiRec [4] further adopts self-attention mechanisms to learn multiple interest representations and dynamically selects the most relevant interest vector during recommendation. SINE [15] models user interests activated in different contexts by constructing sparse interest representations, thereby improving sequential recommendation performance. MIRN [29] leverages multi-interest routing mechanisms to extract multiple interest representations from user behavior sequences, enhancing model expressiveness.

Although these methods have achieved progress in multi-interest modeling, most of them are still based on fine-grained point representations. Such approaches are sensitive to noisy or anomalous data under sparse conditions, making it difficult to learn robust interest structures. Recently, granular-ball computing [25] has provided a new perspective for multi-granularity data modeling. By employing adaptive hyperspheres to cover the data space, this approach captures the structural distribution of data at different granularity levels. Compared with traditional point-based representations, granular-ball computing can preserve structural information while improving robustness to noise, offering a promising direction for multi-granularity interest modeling in recommender systems.

2.3. Graph Neural Network-based Recommendation Methods

Graph neural networks (GNNs) have been widely applied in recommender systems due to their ability to model complex high-order relationships between users and items. By representing user-item interactions as graph structures, GNNs learn node representations through neighborhood aggregation, thereby capturing collaborative filtering signals. NGCF [17] is one of the earliest models that introduces graph convolutional networks into recommender systems, modeling high-order connectivity through multi-layer propagation on the user-item bipartite graph. Subsequently, LightGCN [5] simplifies traditional GCN by removing feature transformation and nonlinear activation, retaining only neighborhood aggregation, which significantly improves recommendation performance while reducing model complexity.

Building upon these methods, subsequent studies have explored more advanced graph modeling techniques. KGAT [16] incorporates knowledge graphs to enrich user-item relationship modeling with additional semantic information. DGCF [19] disentangles graph convolution to capture user-item interactions under different latent intents. SGL [22] enhances representation robustness through graph data augmentation and contrastive learning. SimGCL [28] further improves graph representation learning using a simple yet effective contrastive learning strategy while maintaining model simplicity. LightGCL [3] enhances representation consistency and robustness via graph contrastive learning. DiffRec [20] leverages diffusion models to reconstruct user-item interactions, thereby improving recommendation performance.

Despite these advances, most graph-based recommendation methods primarily rely on user-item interaction structures and lack sufficient modeling of the semantic information underlying user interests. Therefore, integrating semantic enhancement with graph-based learning to construct more accurate and robust recommendation models remains an important research direction.

3. Method

3.1. Problem Definition

Let $\mathcal{U} = \{u_1, u_2, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{I} = \{i_1, i_2, \dots, i_{|\mathcal{I}|}\}$ denote the sets of users and items, respectively. The user-item interaction data is represented as a set $\mathcal{R} \subseteq \mathcal{U} \times \mathcal{I}$, where $(u, i) \in \mathcal{R}$ indicates that user u has interacted with item i . Accordingly, the interaction data can be modeled as a user-item bipartite graph:

$$\mathcal{G}_{ui} = (\mathcal{U} \cup \mathcal{I}, \mathcal{R}). \quad (1)$$

For each user $u \in \mathcal{U}$, the historical interaction sequence is denoted as $\mathcal{H}_u = [i_1, i_2, \dots, i_{n_u}]$, where n_u represents the number of interactions of user u . Since each item is typically associated with textual attributes such as titles or descriptions, let t_i denote the textual information of item i . The textual history of user u can then be defined as $\mathcal{T}_u = [t_{i_1}, t_{i_2}, \dots, t_{i_{n_u}}]$. Based on the above definitions, the goal of this work is to learn representations of users and items from both interaction data and textual information, and to predict user preferences over unobserved items:

$$\hat{y}_{ui} = f(u, i), \quad (2)$$

where $\hat{y}_{ui}$ denotes the predicted preference score of user u for item i , and $f(\cdot)$ represents the proposed recommendation model. The predicted scores are then used to rank candidate items and generate Top- N recommendation results for each user.

3.2. Overall Framework

The overall architecture of the proposed model is illustrated in Fig. 1, which consists of three core components: (1) a user interest generation module, (2) a granular-ball-based interest modeling module, and (3) a graph neural network-based recommendation learning module. These components collaboratively enhance recommendation performance from three perspectives, namely semantic modeling, multi-granularity interest representation, and graph structure learning.

First, in the user interest generation module (Fig. 1(a)), collaborative signals are extracted from the user-item interaction graph $\mathcal{G}_{ui}$, and potential collaborative neighboring items are identified for each user. Based on this, the textual information from user historical interactions and their collaborative neighboring items is jointly fed into a LLM to generate explicit semantic representations of user interests, thereby alleviating the semantic deficiency of traditional ID-based representations. Second, in the granular-ball-based interest modeling module (Fig. 1(b)), the semantic interest representations generated by LLMs are transformed into a vector space, and granular-ball computing is employed to perform multi-granularity interest modeling. Specifically, fine-grained interest phrases are aggregated into structured high-level interest units. Based on this, a user-interest graph is further constructed to capture the intrinsic relationships among user interests and enhance representation robustness. Finally, in the graph neural network-based recommendation

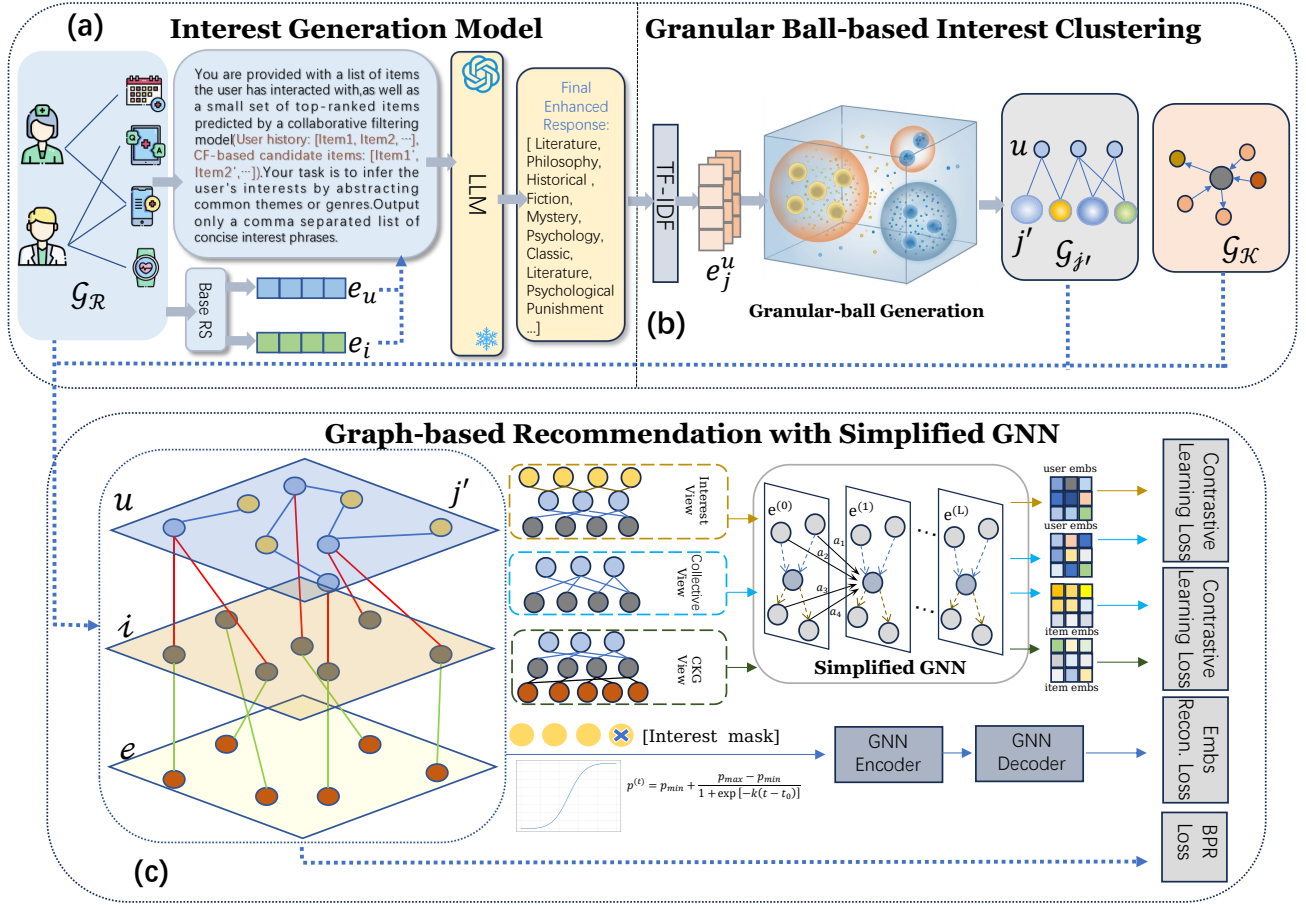


Figure 1: The overall architecture of LLM-GRec.

learning module (Fig. 1(c)), user-item interaction information, interest representations, and structured semantic information are integrated to construct a unified multi-view graph representation, including the interaction view, interest view, and knowledge graph view. Within this framework, a simplified graph neural network is adopted for representation propagation and learning. Combined with contrastive learning and ranking optimization objectives, the model learns discriminative and robust representations of users and items, thereby achieving high-quality personalized recommendation.

3.3. Interest Generation Model

As illustrated in Fig. 1(a), we design an Interest Generation Model, which integrates collaborative filtering signals with textual semantic information to generate explicit semantic representations of user interests via a LLM.

3.3.1. Collaborative Signal Extraction

To capture potential collaborative relationships from user-item interaction data, we first learn user and item representations based on the user-item bipartite graph G_{ui} . Specifically, we construct a normalized adjacency matrix $\hat{A}$ and adopt a lightweight graph propagation mechanism to

update node representations:

$$\mathbf{H}^{(l+1)} = \hat{\mathbf{A}}\mathbf{H}^{(l)}, \quad (3)$$

where $\mathbf{H}^{(l)}$ denotes the node representations at the l -th layer, and $\mathbf{H}^{(0)}$ represents the initial embeddings. After L layers of propagation, the final representation is obtained via layer-wise aggregation:

$$\mathbf{H} = \text{Aggregate}(\mathbf{H}^{(0)}, \mathbf{H}^{(1)}, \dots, \mathbf{H}^{(L)}), \quad (4)$$

where $\text{Aggregate}(\cdot)$ denotes the fusion operation over representations from multiple layers.

The resulting representation $\mathbf{H}$ is then split according to node types to obtain user representations $\mathbf{h}_u$ and item representations $\mathbf{h}_i$. Based on these representations, the similarity between a user and candidate items is computed to capture potential interests:

$$s(u, i) = \text{sim}(\mathbf{h}_u, \mathbf{h}_i). \quad (5)$$

Furthermore, we select the Top- K items with the highest similarity scores from the set of unobserved items to form the collaborative neighbor set:

$$C_u = \text{TopK}(\{s(u, i) \mid i \notin I_u^+\}), \quad (6)$$

where $\mathcal{I}_u^+$ denotes the set of items that user u has previously interacted with. The collaborative neighbor set C_u supplements potential user interests from a structural perspective, providing richer contextual signals for subsequent semantic reasoning by LLMs. This is particularly beneficial in sparse data scenarios, where it effectively alleviates the issue of insufficient interest information.

3.3.2. LLM-based Semantic Interest Generation

After obtaining the collaborative neighbor set C_u , we further leverage a LLM to perform semantic modeling of user interests, thereby compensating for the limitation of traditional collaborative filtering methods in semantic representation. Specifically, for each user u , the input to the LLM consists of two components. The first is the user's historical interaction text sequence $\mathcal{T}_u$, which reflects explicit user preferences. The second is the textual set corresponding to collaborative neighboring items, denoted as $\mathcal{T}_u^{cf} = [t_{c_1}, t_{c_2}, \dots, t_{c_K}]$, where $c_k \in C_u$, which serves to supplement potential interest information. Based on this, the two types of textual information are jointly constructed as a prompt and fed into the LLM for semantic reasoning:

$$\mathbf{z}_u = \text{LLM}(\mathcal{T}_u, \mathcal{T}_u^{cf}), \quad (7)$$

where $\mathbf{z}_u$ denotes the semantic representation of user interests. Unlike approaches that rely solely on user historical behaviors, the incorporation of collaborative neighbor information provides structured supplementary signals for LLMs, enabling them to generate more comprehensive and stable semantic representations, especially under sparse interaction scenarios. Furthermore, the semantic representation generated by the LLM is parsed into a set of interest keywords or phrases, i.e., $\mathbf{z}_u = \{w_1, w_2, \dots, w_{K_u}\}$, where w_k denotes an interest keyword or topic label, and K_u represents the number of extracted interests. This representation models user preferences at the natural language level, offering stronger semantic expressiveness and interpretability compared to traditional ID-based representations, and providing a solid foundation for subsequent multi-granularity interest modeling.

3.4. Granular Ball-based Interest Clustering

To extract more robust and structured semantic representations from fine-grained interest phrases generated by LLMs, we propose a **Granular Ball-based Interest Clustering** module, as illustrated in Fig. 1(b). Since LLM-generated interests are typically represented as high-dimensional and sparse phrase sets that may contain redundancy and noise, directly modeling them with point-wise representations is susceptible to noisy or anomalous data. To address this issue, we introduce granular-ball computing [24], which constructs adaptive granular-ball structures to aggregate fine-grained interests into multi-granularity high-level interest units, thereby improving representation robustness.

3.4.1. Interest Phrase Representation and Local Semantic Modeling

Based on the semantic interest representation generated by LLMs in the previous section, $\mathbf{z}_u = \{w_1, w_2, \dots, w_{K_u}\}$, each keyword or phrase is treated as an individual interest unit. Accordingly, we construct the interest phrase set for user u as $\mathcal{P}_u = \{p_1^u, p_2^u, \dots, p_{m_u}^u\}$. To capture semantic relationships among interest phrases, we encode each phrase using the TF-IDF scheme, yielding vector representations:

$$\mathbf{x}_j^u = \text{TFIDF}(p_j^u), \quad (8)$$

where $\mathbf{x}_j^u \in \mathbb{R}^d$ denotes the vector representation of the j -th interest phrase of user u . Based on these representations, we construct local semantic associations between interest phrases using cosine similarity:

$$s(\mathbf{x}_j^u, \mathbf{x}_k^v) = \frac{\mathbf{x}_j^u}{\|\mathbf{x}_j^u\|} \cdot \frac{\mathbf{x}_k^v}{\|\mathbf{x}_k^v\|}. \quad (9)$$

Subsequently, for each interest phrase, only the Top- K most similar neighbors are retained to construct its local semantic neighborhood. This local neighborhood structure serves as the foundation for subsequent granular-ball construction, enabling semantically similar phrases to be aggregated into the same high-level interest units during multi-granularity modeling.

3.4.2. Granular Ball Construction and Interest Graph Modeling

After obtaining the vector representations of interest phrases $\mathbf{x}_j^u$, we further introduce granular-ball computing to perform multi-granularity interest modeling, aiming to extract more stable high-level semantic structures from fine-grained interest phrases. Specifically, based on the local semantic associations among interest phrases, semantically similar phrase vectors are aggregated into multiple granular balls, each representing a high-level interest unit. Formally, a granular ball can be defined as a set of semantically similar phrase vectors, i.e., $\mathcal{B} = \{\mathbf{x}_{j_1}, \mathbf{x}_{j_2}, \dots, \mathbf{x}_{j_n}\}$, where the vectors within $\mathcal{B}$ exhibit high semantic consistency in the embedding space. Each granular ball is treated as an abstract interest node j' , which captures a user's interest pattern at a specific semantic granularity. To obtain high-quality granular-ball partitions, we adopt a quality-driven recursive partitioning strategy. For any granular ball $\mathcal{B}$, its quality is defined as:

$$Q(\mathcal{B}) = \frac{2|\mathcal{E}_{\mathcal{B}}|}{|\mathcal{V}_{\mathcal{B}}|} + \text{Transitivity}(\mathcal{B}), \quad (10)$$

where $\mathcal{V}_{\mathcal{B}}$ and $\mathcal{E}_{\mathcal{B}}$ denote the sets of nodes and edges within the granular ball, respectively. This quality function jointly characterizes the internal connectivity density and local structural consistency of the granular ball. During the partitioning process, an initial granular ball is first constructed from the interest phrase set, and then recursively split into two sub-balls $\mathcal{B}_1$ and $\mathcal{B}_2$ according to the following criterion:

$$\text{Split}(\mathcal{B}) = \begin{cases} \text{True,} & \text{if } Q(\mathcal{B}_1) + Q(\mathcal{B}_2) > Q(\mathcal{B}) \\ \text{False,} & \text{otherwise.} \end{cases} \quad (11)$$

The above process is repeated until convergence of the quality measure or a minimum granularity constraint is satisfied. Through this procedure, each granular ball corresponds to a stable high-level interest unit, effectively mapping fine-grained interest phrases to coarse-grained semantic representations. Based on the resulting granular-ball partition, we further construct a user-interest graph. Specifically, if an interest phrase $\mathbf{x}_j^u$ of user u belongs to a granular ball j' , an edge is established between the user node and the corresponding interest node. The resulting user-interest graph is defined as:

$$\mathcal{G}_{uj} = (\mathcal{U} \cup \mathcal{J}', \mathcal{E}_{uj}), \quad (12)$$

where $\mathcal{E}_{uj} = \{(u, j') \mid \exists \mathbf{x}_j^u \in j'\}$. Furthermore, to incorporate richer structured semantic information, we additionally consider the item-side knowledge graph $\mathcal{G}_k$, which is jointly modeled with the user-interest graph in subsequent modules to enhance representation learning.

3.5. Graph-based Recommendation with Simplified GNN

As illustrated in Fig. 1(c), after constructing a unified graph that incorporates users, items, entities, and granular-ball interest nodes, we design a multi-view learning framework based on a **Simplified GNN** to achieve efficient and robust recommendation.

3.5.1. Multi-view Graph Construction

To effectively integrate collaborative signals, semantic interests, and structured knowledge, we construct a unified multi-view graph representation framework. In this framework, different types of relations are organized into multiple views, enabling collaborative modeling while preserving semantic independence. We first define a unified node set:

$$\mathcal{V} = \mathcal{U} \cup \mathcal{I} \cup \mathcal{J}' \cup \mathcal{E}, \quad (13)$$

where $\mathcal{J}'$ denotes the set of granular-ball interest nodes, and $\mathcal{E}$ denotes the set of entity nodes in the knowledge graph. Based on this, we construct three key views:

(1) Interaction view. This view captures explicit user-item interaction relationships:

$$\mathcal{G}_{cf} = (\mathcal{V}, \mathcal{E}_{cf}), \quad (14)$$

where $\mathcal{E}_{cf}$ denotes the user-item interaction edges. This view provides the fundamental collaborative filtering signals.

(2) Interest view. Based on the granular-ball interest modeling results, we introduce user-interest connections:

$$\mathcal{G}_{ig} = (\mathcal{V}, \mathcal{E}_{cf} \cup \mathcal{E}_{uj}), \quad (15)$$

where $\mathcal{E}_{uj}$ denotes the edges between users and interest nodes. This view incorporates high-level semantic preferences into the graph structure, helping to alleviate data sparsity and enhance semantic expressiveness.

(3) Knowledge graph view. To model structured semantic relationships among items, we incorporate knowledge graph edges:

$$\mathcal{G}_{kg} = (\mathcal{V}, \mathcal{E}_{cf} \cup \mathcal{E}_k), \quad (16)$$

where $\mathcal{E}_k$ denotes the relational edges in the knowledge graph. This view provides rich structured semantic information, complementing the semantic limitations of interaction-only data. In the above multi-view structure, all views share a unified node embedding space, while different views impose constraints and propagate information through their respective edge structures. In this way, collaborative, semantic, and structural information are jointly modeled in a unified framework.

3.5.2. Dynamic Mask-based Interest Denoising Mechanism

Although LLMs can generate semantically rich user interest representations, their outputs inevitably contain a certain degree of noise and bias. Directly incorporating such interest nodes into graph propagation may introduce erroneous semantics, which can be further amplified through multi-hop message passing, thereby degrading model stability and recommendation performance. Recent studies in self-supervised recommendation have demonstrated that masking, augmentation, and reconstruction strategies play a crucial role in improving representation quality [14]. Inspired by curriculum learning and dynamic masking strategies in graph masked autoencoders [2, 21], we adopt an S-shaped scheduling function to dynamically adjust the masking rate of interest nodes. This strategy allows the model to maintain low perturbation in early training stages for stable representation learning, and gradually increase denoising strength in later stages to enhance robustness.

To avoid the limitations of fixed masking strategies, which may lead to either excessive information loss or insufficient denoising, we design a dynamically adaptive masking rate that evolves with the training process. Specifically, the masking rate at the t -th training iteration is defined as:

$$p^{(t)} = p_{\min} + \frac{p_{\max} - p_{\min}}{1 + \exp(-k(t - t_0))}, \quad (17)$$

where $p_{\min}$ and $p_{\max}$ denote the minimum and maximum masking rates, respectively, k controls the slope of the function, and t_0 represents the midpoint of the transition. This S-shaped function ensures that the model starts with low perturbation to stabilize learning, and gradually increases the denoising intensity as training progresses. Given the masking rate $p^{(t)}$, we apply stochastic perturbations to the interest node representations. Let $\mathbf{X}^{\mathcal{J}'}$ denote the initial embeddings of interest nodes. The perturbed representations are obtained as:

$$\tilde{\mathbf{X}}^{\mathcal{J}'} = \mathcal{M}(\mathbf{X}^{\mathcal{J}'}, p^{(t)}), \quad (18)$$

where $\mathcal{M}(\cdot)$ denotes the masking operation, which randomly selects a subset of interest nodes and applies zero-masking or noise injection. This process simulates uncertainty in interest representations and enhances the model's robustness to noisy signals. To further improve the stability of learned representations, we adopt an encoder-decoder architecture to reconstruct the perturbed interest nodes. Specifically, the perturbed node features and graph structure are first encoded, followed by a decoding process to recover the original representations:

$$\mathbf{H}_{enc} = f_{enc}(\tilde{\mathbf{X}}, \tilde{\mathcal{G}}), \quad \mathbf{H}_{dec} = f_{dec}(\mathbf{H}_{enc}, \tilde{\mathcal{G}}), \quad (19)$$

where $\tilde{\mathcal{G}}$ denotes the graph with randomly perturbed edges, $\mathbf{h}_i^{init}$ represents the original embedding, and $\mathbf{h}_i^{dec}$ denotes the reconstructed embedding. For the masked interest nodes, we employ a similarity-based reconstruction loss:

$$\mathcal{L}_{rec} = \frac{1}{|M|} \sum_{i \in M} (1 - \cos(\mathbf{h}_i^{dec}, \mathbf{h}_i^{init}))^\alpha, \quad (20)$$

where M denotes the set of masked nodes and α is a hyperparameter controlling the loss sensitivity. Through this perturbation-reconstruction paradigm, the model is encouraged to distinguish between noisy and informative semantic signals during the reconstruction process, thereby mitigating the impact of spurious interests in graph propagation. Moreover, the dynamic S-shaped masking strategy prevents performance degradation caused by overly strong or weak perturbations, ensuring stable and effective learning throughout different training stages. Overall, the proposed denoising mechanism enables the model to suppress potential noise introduced by LLM-generated interests while preserving semantic expressiveness, thus providing more reliable inputs for graph-based representation learning.

3.5.3. Representation Learning and Optimization with Simplified GNN

Given the unified initial node representations $\mathbf{H}^{(0)}$, we perform representation learning over the multi-view graph using a simplified graph convolution mechanism. The update rule at the l -th layer is defined as:

$$\mathbf{H}^{(l+1)} = \gamma \cdot \text{GNN}(\mathbf{H}^{(l)}, \mathcal{G}) + \eta \cdot \mathbf{H}^{(l)}, \quad (21)$$

where γ is the global aggregation weight, η is the residual coefficient, and $\text{GNN}(\cdot)$ denotes the neighbor aggregation operation based on the normalized adjacency matrix. After L propagation layers, the final node representations are obtained via layer-wise averaging:

$$\mathbf{H} = \frac{1}{L+1} \sum_{l=0}^L \mathbf{H}^{(l)}. \quad (22)$$

Under different views, the learned representations are denoted as $\mathbf{H}_{cf}$, $\mathbf{H}_{ig}$, and $\mathbf{H}_{kg}$, corresponding to the interaction view, interest view, and knowledge graph view, respectively. To enhance consistency across different semantic spaces, we adopt cross-view contrastive learning for representation

alignment. Specifically, for user nodes, we align representations from the interaction view and the interest view; for item nodes, we align representations from the interaction view and the knowledge graph view. The overall contrastive learning objective is defined as:

$$\mathcal{L}_{cl} = \mathcal{L}_{cl}^u + \mathcal{L}_{cl}^i. \quad (23)$$

Taking user nodes as an example, the contrastive loss is formulated as:

$$\mathcal{L}_{cl}^u = - \sum_u \log \frac{\exp(\text{sim}(\mathbf{h}_{cf}^u, \mathbf{h}_{ig}^u)/\tau)}{\sum_v \exp(\text{sim}(\mathbf{h}_{cf}^u, \mathbf{h}_{ig}^v)/\tau)}, \quad (24)$$

where $\text{sim}(\cdot)$ denotes the similarity function and τ is a temperature parameter that controls the smoothness of the similarity distribution. The contrastive loss for item nodes $\mathcal{L}_{cl}^i$ is defined in a similar manner. For the recommendation task, we adopt the Bayesian Personalized Ranking (BPR) loss to model user preferences:

$$\mathcal{L}_{bpr} = - \sum_{(u,i,j)} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}), \quad (25)$$

where (u, i, j) denotes a triplet in which user u prefers item i over item j . Finally, the overall optimization objective integrates the recommendation loss, contrastive learning loss, and reconstruction loss as follows:

$$\mathcal{L} = \lambda_{bpr} \mathcal{L}_{bpr} + \lambda_{cl} \mathcal{L}_{cl} + \lambda_{rec} \mathcal{L}_{rec} + \lambda_{reg} \|\Theta\|_2^2, \quad (26)$$

where Θ denotes the set of model parameters and λ_* are trade-off hyperparameters. Through this joint optimization, the model aligns representations across multiple views while suppressing noisy interest signals, thereby learning more robust and discriminative user and item embeddings for recommendation.

4. Experiments

To comprehensively evaluate the effectiveness and robustness of the proposed **LLM-GRec**, we conduct extensive experiments on three real-world datasets. The evaluation is organized around the following research questions:

- **RQ1:** Can LLM-GRec achieve superior recommendation performance compared with state-of-the-art methods?
- **RQ2:** Do all key components of the model contribute to the final performance?
- **RQ3:** How sensitive is the model to critical hyperparameters, and does it exhibit stable performance?
- **RQ4:** Is LLM-GRec robust under scenarios with semantic noise?
- **RQ5:** Are the learned interest representations interpretable and reflective of users' true preferences?

Table 1

Performance comparison results on three datasets. The best results are shown in bold, and the second-best results are underlined. “%Imp.” denotes the percentage improvement of LLM-GRec compared to the second-best results

MODEL	DBbook2014				Book-Crossing				MovieLens-1M			
	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑
LightGCN[5](SIGIR20)	0.4214	0.5358	0.2230	0.2481	0.1607	0.2246	0.0765	0.0911	0.4086	0.5574	0.3977	0.4432
SGL[22](SIGIR21)	0.4258	0.5249	0.2350	0.2567	0.1613	0.2154	0.0860	0.0982	0.4094	0.5505	0.4004	0.4433
SimGCL[28](SIGIR22)	0.4280	0.5369	0.2376	0.2610	0.1653	0.2211	0.0860	0.0988	0.4131	0.5560	0.4048	0.4482
CKE[31](KDD16)	0.3419	0.4448	0.1832	0.2060	0.1284	0.1691	0.0708	0.0811	0.3948	0.5421	0.3841	0.4295
CFKG[1](AI2018)	0.3586	0.4559	0.1627	0.1816	0.1216	0.1771	0.0503	0.0617	0.3930	0.5439	0.3672	0.4142
KGAT[16](KDD19)	0.4174	0.5309	0.2110	0.2360	0.1564	0.2133	0.0865	0.1004	0.4059	0.5527	0.3928	0.4378
KGIN[18](WWW21)	0.4147	0.5337	0.2200	0.2462	0.1468	0.2075	0.0681	0.0819	0.4095	0.5581	0.4004	0.4453
KGCL[26](SIGIR22)	0.4308	0.5271	0.2453	0.2666	0.1562	0.2049	0.0859	0.0971	0.4083	0.5496	0.3996	0.4430
MCCLK[32](SIGIR22)	0.4376	0.5491	0.2054	0.2273	0.1476	0.2039	0.0656	0.0774	0.4176	0.5655	0.3876	0.4339
KGRec[27](KDD23)	0.4415	0.5497	0.2373	0.2611	0.1473	0.2075	0.0699	0.0836	0.4136	0.5633	0.4072	0.4524
CIKGRec[6](AAAI25)	0.4642	0.5756	0.2494	0.2739	0.1791	0.2455	0.0889	0.1041	0.4250	0.5718	0.4162	0.4609
LLM-GRec	0.4671	<u>0.5741</u>	0.2576	0.2798	0.1866	0.2501	0.0967	0.1107	0.4341	0.5806	0.4230	0.4680
%Imp.	0.62%	-0.26%	3.29%	2.15%	4.18%	1.87%	8.78%	6.34%	2.14%	1.54%	1.63%	1.55%

Table 2

Statistics of Datasets

Dataset	DBbook2014	Book-Crossing	MovieLens-1M
#Users	5,576	6,616	6,040
#Items	2,680	8,853	3,260
#Interactions	65,961	110,662	998,539
#Density	0.44%	0.19%	5.07%
#Relations	13	4	20
#Entities	8,762	1,404	14,377
#Triplets	134,223	1,137	415,104

- **RQ6:** Does LLM-GRec incur significant computational overhead while achieving superior performance?
- **RQ7:** How does LLM-GRec perform compared with other models under sparse interaction scenarios?

4.1. Experimental Settings

4.1.1. Datasets

To comprehensively evaluate the proposed model, we conduct experiments on three widely used real-world recommendation datasets: **DBbook2014**, **Book-Crossing**, and **MovieLens-1M**. These datasets differ significantly in terms of user scale, interaction sparsity, and richness of semantic information, allowing us to validate the effectiveness and generalization ability of the model from multiple perspectives. For all datasets, we preprocess the user-item interaction data by converting explicit ratings into implicit feedback. Following standard practice, the data is split into training, validation, and test sets. Detailed statistics of the datasets are reported in Table 2.

4.1.2. Baselines

To comprehensively evaluate the performance of the proposed model, we compare it with a set of representative recommendation methods. According to the utilized information sources and modeling paradigms, these methods are categorized into three groups:

(1) **Graph-based collaborative filtering methods:** These methods model high-order collaborative signals based on the user-item interaction graph.

- **LightGCN** [5]: Simplifies graph convolution by retaining only neighborhood aggregation, enabling efficient collaborative filtering.
- **SGL** [22]: Introduces self-supervised learning with graph augmentation and contrastive learning to enhance representation robustness.
- **SimGCL** [28]: Enhances generalization ability via simple noise injection and contrastive learning.

(2) **Knowledge graph-enhanced recommendation methods:** These methods incorporate external knowledge graphs to model semantic relationships among items, alleviating the limitations of interaction-only data.

- **CKE** [31]: Jointly learns collaborative filtering signals and knowledge graph embeddings.
- **CFKG** [1]: Unifies collaborative filtering and knowledge graph information into a single graph for representation learning.
- **KGAT** [16]: Integrates interaction data and knowledge graph information via graph attention mechanisms.
- **KGIN** [18]: Models user preferences under different semantic relations through intent disentanglement.
- **KGCL** [26]: Combines knowledge graph modeling with contrastive learning to improve representation consistency.
- **MCCLK** [32]: Employs multi-view contrastive learning to jointly model collaborative and knowledge signals.
- **KGRec** [27]: Enhances recommendation performance via knowledge-aware representation learning.

(3) LLM-enhanced recommendation methods: These methods leverage LLM to improve semantic representation learning.

- **CIKGRec** [6]: Integrates LLM-generated semantic information with a knowledge-enhanced recommendation framework.

4.1.3. Evaluation Metrics

We evaluate model performance using widely adopted metrics for Top- N recommendation, including **Recall@K** and **NDCG@K**, where $K \in \{50, 100\}$. **Recall@K** measures the ability of the model to retrieve relevant items from the candidate set, defined as the proportion of ground-truth items appearing in the top- K recommendation list. This metric emphasizes the recall capability of the model. **NDCG@K** evaluates the ranking quality of the recommendation results. Unlike Recall, NDCG considers the positions of items in the ranked list and assigns higher weights to correctly recommended items at top positions, thus providing a more comprehensive assessment of ranking performance. In all experiments, candidate items are ranked based on predicted scores. Recall@K and NDCG@K are computed for each user and then averaged across all users.

4.1.4. Implementation Details

To ensure fair comparison, we adopt consistent experimental settings across all methods. The embedding dimension is set to $d = 128$, and the number of GNN layers is $L = 3$. We train the model using the Adam optimizer with a learning rate of 5×10^{-4} , batch size of 512, and a maximum of 120 training epochs. Early stopping is applied based on validation performance. For the contrastive learning module, the temperature parameter is set to 0.15, and the corresponding loss weight is $\lambda_{cl} = 0.02$. For the interest denoising and reconstruction module, the reconstruction loss weight is $\lambda_{rec} = 0.05$, and a similarity-based consistency loss is adopted with exponent $\alpha = 2$. For the dynamic masking mechanism, the minimum and maximum masking rates are set to $p_{min} = 0.1$ and $p_{max} = 0.5$, respectively, and are progressively adjusted during training. The encoder and decoder are configured with 2 and 1 layers, respectively, to effectively reconstruct interest representations. For the LLM module, we utilize a pre-trained LLM to generate user interest keywords in an offline manner. Specifically, user historical interaction texts and collaborative neighbor item texts are constructed as prompts to generate structured interest representations, which remain fixed during subsequent training. All models are implemented under the same experimental environment, and hyperparameters are tuned based on the validation set to ensure fairness and reproducibility.

4.2. Overall Performance Comparison (RQ1)

We compare the proposed LLM-GRec with a range of state-of-the-art recommendation methods on three real-world datasets. The results are reported in Table 1. Overall,

LLM-GRec achieves the best performance on most evaluation metrics across all datasets, demonstrating its effectiveness in semantic modeling and robust interest learning.

First, from a general perspective, methods that incorporate semantic information consistently outperform those relying solely on collaborative signals. For instance, models such as LightGCN, SGL, and SimGCL are purely based on user-item interactions, whereas approaches that integrate knowledge graphs or semantic information (e.g., KGAT, KGIN, and CIKGRec) generally achieve superior performance. This observation indicates that additional semantic signals provide complementary information beyond collaborative interactions, thereby alleviating data sparsity and improving recommendation accuracy.

Second, among semantic-enhanced approaches, methods leveraging LLMs exhibit stronger performance potential. For example, CIKGRec, which incorporates LLM-generated semantic information, already surpasses traditional graph-based recommendation models. However, such methods remain vulnerable to the noise introduced by LLM-generated content, which limits their performance. In contrast, LLM-GRec effectively mitigates this issue through multi-granularity interest modeling and a dynamic denoising mechanism, enabling the model to learn more stable and discriminative user representations.

Finally, LLM-GRec consistently achieves the best results on the majority of metrics across all datasets and significantly outperforms the strongest baseline (CIKGRec) on several metrics. For example, on the Book-Crossing dataset, LLM-GRec improves NDCG@50 by 8.78%. It also achieves consistent performance gains on DBbook2014 and MovieLens-1M. These results demonstrate the strong generalization ability of the proposed method under different data distributions. Notably, the advantage of LLM-GRec becomes more pronounced in sparse interaction scenarios. This can be attributed to the complementary semantic signals provided by LLMs, as well as the synergy between multi-granularity interest modeling and the proposed denoising mechanism, which together significantly enhance both robustness and recommendation accuracy.

4.3. Ablation Study (RQ2)

We conduct ablation studies to analyze the contributions of key components in the proposed model. Specifically, based on the full model LLM-GRec, we progressively remove or replace different modules to evaluate their impact on recommendation performance.

According to the model design, we focus on the following core components:

- **w/o LLM:** Removes the semantic interest representations generated by the LLM, and relies solely on the user-item interaction graph. This variant is used to evaluate the contribution of semantic information to recommendation performance.
- **w/o CF-guided:** During the interest generation stage, only user historical interaction texts are used, without

Table 3

Ablation study results of different LLM-GRec variants on DBbook2014, Book-Crossing, and MovieLens-1M datasets.

Model	DBbook2014				Book-Crossing				MovieLens-1M			
	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑	R@50 ↑	R@100 ↑	N@50 ↑	N@100 ↑
LLM-GRec	0.4671	0.5741	0.2576	0.2798	0.1866	0.2501	0.0967	0.1107	0.4341	0.5806	0.4230	0.4680
w/o LLM	0.4374	0.5532	0.2357	0.2610	0.1701	0.2340	0.0831	0.0977	0.4091	0.5551	0.4022	0.4462
w/o CF-guided	0.4668	0.5701	0.2556	0.2788	0.1850	0.2485	0.0962	0.1108	0.4315	0.5786	0.4200	0.4657
w/o GB	0.4432	0.5518	0.2392	0.2631	0.1856	0.2476	0.0942	0.1085	0.4323	0.5780	0.4195	0.4650
w/o Denoising	0.4662	0.5722	0.2529	0.2763	0.1851	0.2482	0.0937	0.1082	0.4261	0.5734	0.4160	0.4610
w/o CL	0.4565	0.5681	0.2470	0.2715	0.1683	0.2349	0.0831	0.0980	0.4058	0.5527	0.3995	0.4440
w/o KG View	0.4544	0.5641	0.2490	0.2730	0.1686	0.2339	0.0820	0.0969	0.4093	0.5712	0.4102	0.4518

incorporating collaborative neighbor items. This setting is designed to analyze the role of collaborative signals in semantic modeling.

- **w/o GB**: Replaces the granular-ball clustering module with K-means clustering, in order to verify the effectiveness of multi-granularity interest modeling for improving robustness.
- **w/o Denoising**: Removes the dynamic mask-based denoising and reconstruction mechanism, to investigate its effectiveness in mitigating semantic noise.
- **w/o CL**: Removes the cross-view contrastive learning loss and retains only the basic recommendation objective, to evaluate the role of contrastive learning in enhancing representation consistency.
- **w/o KG View**: Removes the knowledge graph view and performs modeling only based on the interaction graph and interest graph, to analyze the contribution of structured semantic information.

The results are reported in Table 3. We observe that removing any core component from LLM-GRec generally leads to performance degradation across most datasets, highlighting the effectiveness of each module in the proposed framework. Specifically, removing the LLM-based semantic interest generation module (w/o LLM) results in a significant performance drop on all datasets, with more pronounced degradation on sparse datasets such as Book-Crossing. This indicates that the semantic interests generated by LLMs play a crucial role in alleviating data sparsity and enhancing recommendation performance. When removing the collaborative neighbor guidance (w/o CF-guided), the performance shows a moderate decline. This suggests that collaborative signals provide complementary semantic cues for LLMs, thereby improving the completeness and accuracy of interest modeling. Replacing the granular-ball interest modeling module with K-means clustering (w/o GB) leads to a substantial performance drop. This demonstrates that the proposed multi-granularity interest representation is more effective in capturing structured user interests and enhances robustness in complex semantic spaces. Removing the denoising mechanism (w/o Denoising) consistently degrades performance across datasets, indicating that the dynamic masking and reconstruction strategy effectively suppresses

semantic noise and improves the reliability of learned representations. Eliminating the contrastive learning module (w/o CL) results in noticeable performance decline, suggesting that cross-view contrastive learning helps align representations and improves their discriminative power. Finally, removing the knowledge graph view (w/o KG View) also leads to performance degradation, which indicates that structured semantic information provides valuable complementary signals for item representation learning.

Overall, these results answer RQ2 by demonstrating that each component of LLM-GRec contributes positively to the final performance, and the combination of all modules leads to the best results.

4.4. Parameter Sensitivity Analysis (RQ3)

We further investigate the sensitivity of LLM-GRec to key hyperparameters. Specifically, we examine the impact of the neighborhood size parameter K in the granular-ball modeling module. The experimental results are illustrated in Fig. 2. From the overall trend, the model performance first increases and then decreases as K grows. The optimal performance is achieved at $K = 10$ on the Book-Crossing dataset, while the best result on DBbook2014 is obtained at $K = 12$. Specifically, when K is relatively small, each interest phrase relies on limited neighborhood information, making it difficult to capture sufficient local semantic structure, which leads to suboptimal performance. As K increases, more relevant semantic information is incorporated, enabling the construction of more stable granular-ball structures and thus improving recommendation performance. However, when K becomes too large, excessive irrelevant neighbors are introduced, which injects noise into the interest representations and consequently degrades model performance. These results indicate that there exists a trade-off between semantic completeness and noise introduction when selecting K , and an appropriate neighborhood size is essential for achieving optimal performance.

Furthermore, we investigate the impact of the temperature parameter τ in contrastive learning on model performance. The experimental results are shown in Fig. 3. From the overall trend, the model performance first improves and then slightly declines as τ increases. On both DBbook2014 and Book-Crossing datasets, the optimal performance is achieved when τ is in the range of approximately 0.1–0.15. Specifically, when τ is very small (e.g., $\tau = 0.01$), the contrastive learning objective excessively emphasizes hard

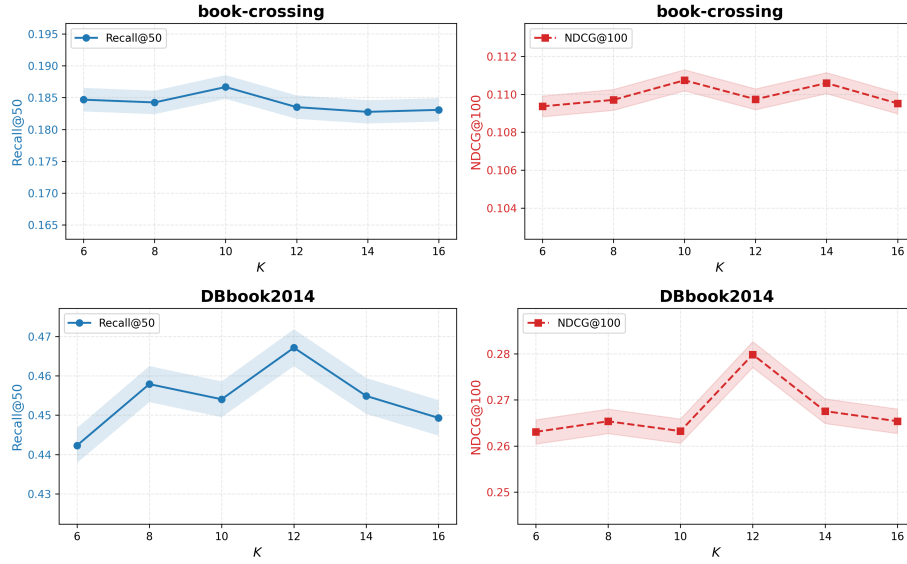


Figure 2: Performance of varying neighborhood sizes in granular ball modeling on Book-Crossing and DBbook2014 datasets.

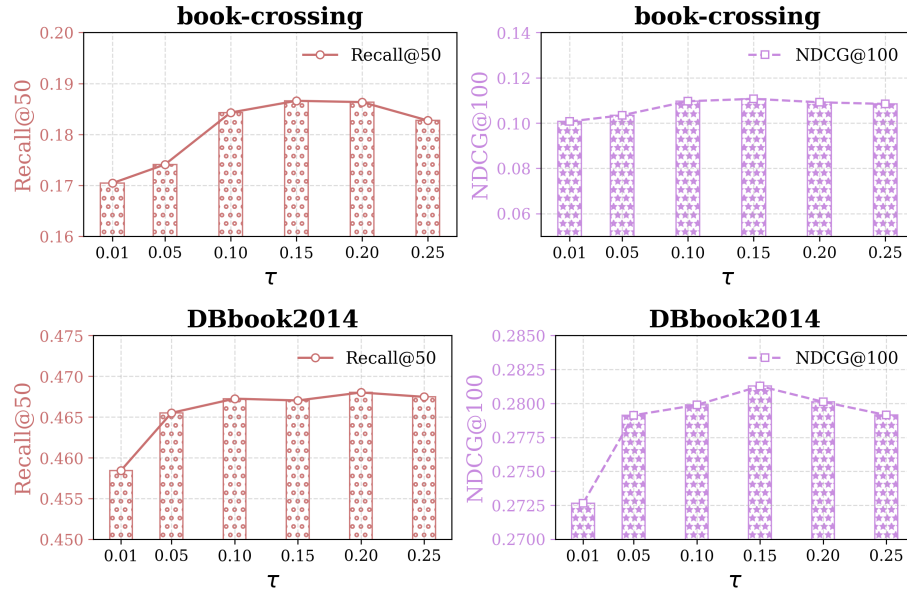


Figure 3: Performance with varying temperature parameters on Book-Crossing and DBbook2014 datasets.

negative samples, which may lead to unstable representations and thus degrade recommendation performance. As τ increases, the model achieves a more balanced separation between positive and negative samples, resulting in improved representation quality. However, when τ becomes too large, the distinction between positive and negative samples weakens, leading to a slight performance decline. In addition, the model maintains relatively stable performance over a wide range of τ values, indicating that the proposed method is robust to the choice of the temperature parameter and does not rely on fine-grained hyperparameter tuning to achieve strong performance.

4.5. Robustness Analysis (RQ4)

In the ablation study, we observe that removing the denoising module (w/o Denoising) does not lead to a substantial performance drop under the original interaction data, possibly because the inherent semantic noise is not sufficiently pronounced. Therefore, we further evaluate the robustness of the proposed model under explicitly injected semantic noise. Specifically, we simulate noisy semantic conditions by randomly injecting noise into the interest phrases generated by the LLM. A certain proportion of interest phrases is replaced with irrelevant or perturbed phrases to mimic semantic deviations. The noise ratio is gradually increased from 0% to 50%, and the results are illustrated in

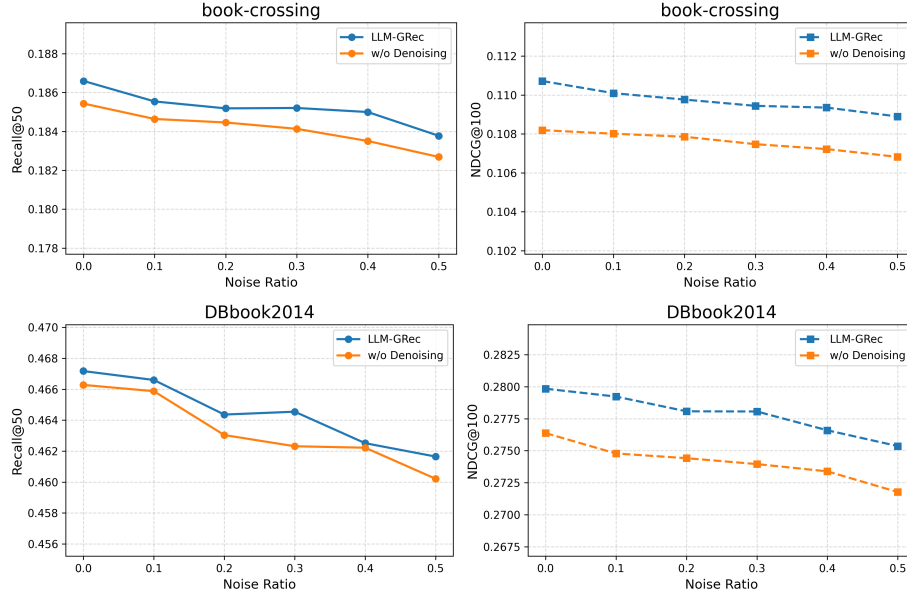


Figure 4: Performance of LLM-GRec and its variant w/o Denoising with varying noise injection ratios on Book-Crossing and DBbook2014 datasets.

Fig. 4. From the overall trend, the performance of all models consistently decreases as the noise ratio increases, indicating that semantic noise negatively impacts recommendation performance. However, compared with the variant without the denoising mechanism (w/o Denoising), LLM-GRec consistently achieves superior performance across all noise levels and exhibits a significantly slower degradation trend. On both Book-Crossing and DBbook2014 datasets, LLM-GRec outperforms w/o Denoising across the entire noise spectrum, with the performance gap becoming more pronounced under high noise levels (e.g., 40%–50%). This demonstrates that the proposed denoising mechanism effectively mitigates the impact of noisy semantic signals and enhances the robustness of the model against unreliable interest information. Moreover, even under high noise ratios, LLM-GRec maintains relatively stable performance, indicating its ability to learn discriminative user representations despite incomplete or corrupted semantic inputs. These results clearly answer RQ4 and validate the effectiveness of the proposed dynamic mask-based denoising strategy.

4.6. Case Study(RQ5)

We present a case study to illustrate the full reasoning process of the proposed model, from user historical behaviors to interest modeling and final recommendation results, as shown in Fig. 5. For user #u6872, the historical interactions are primarily concentrated on medieval-themed mystery novels and the *The Cat Who* series, such as *Brother Cadfael's Penance*, *The Heretic's Apprentice*, and related works. These interactions reflect the user's preference for historical mystery and specific thematic content. Based on this, the model further incorporates collaborative filtering candidate items as supplementary semantic signals and jointly feeds the user history and candidate item texts

into the LLM to generate enriched interest representations. It can be observed that the generated interest themes, including *Historical Fiction*, *Mystery*, *Crime*, *Medieval Themes*, and *Cat Sleuth*, accurately summarize the user's reading preferences and provide a high-level semantic abstraction of their interests. Guided by these semantic representations, the final recommendation results exhibit strong consistency with the inferred interest themes. For example, the recommended list includes multiple works from the *Brother Cadfael* series and *The Cat Who* series, which align closely with both the user's historical interactions and the extracted interest topics. This case study clearly demonstrates the end-to-end reasoning process of LLM-GRec, where user behaviors are transformed into semantic interests via LLM-based abstraction and subsequently guide the recommendation generation. The results indicate that LLM-GRec not only improves recommendation performance but also learns interpretable interest representations, providing intuitive and meaningful semantic explanations for the recommended items.

4.7. Efficiency Analysis (RQ6)

To further evaluate the computational efficiency of the proposed model, we compare the average training time per epoch of different recommendation methods on both a sparse dataset (DBbook2014) and a dense dataset (MovieLens-1M). The results are shown in Fig. 6. Overall, LLM-GRec incurs higher training time compared to LightGCN, KGIN, and CIKGRec on both datasets. This is mainly due to the additional modules introduced in our framework, including interest modeling, multi-view graph propagation, and denoising learning, which inevitably introduce extra computational overhead. However, the increase in training time remains within an acceptable range and does not lead to

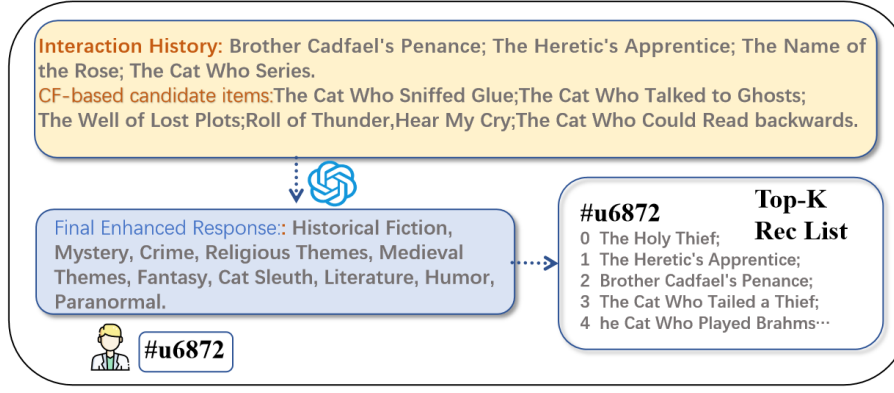


Figure 5: Case Study: Interpretability of User Interest Representation

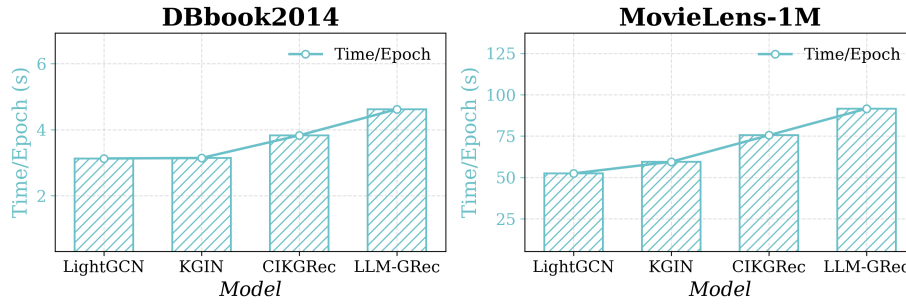


Figure 6: Comparison of running efficiency among different models on DBbook2014 and MovieLens-1M.

significant efficiency degradation. Specifically, on the DBbook2014 dataset, LightGCN and KGIN exhibit comparable training efficiency, while CIKGRc incurs slightly higher computational cost. LLM-GRec shows the highest training time due to its more complex modeling components. On the MovieLens-1M dataset, the overall training time of all methods increases with the data scale, while the relative efficiency ranking among models remains consistent with that observed on DBbook2014. This indicates that although LLM-GRec introduces additional computational overhead, it maintains good scalability as the dataset size grows. It is worth noting that the generation of user interest representations via the LLM is performed offline. Therefore, the training time reported in Fig. 6 mainly reflects the online training cost and does not include the one-time cost of interest generation. In other words, the additional computational overhead of LLM-GRec primarily stems from the interest graph modeling and denoising modules, rather than repeated invocation of the LLM during each training iteration.

4.8. Sparsity Analysis (RQ7)

To further evaluate the performance of the proposed model under different levels of interaction sparsity, we construct datasets with varying sparsity by controlling the data retention ratio $\rho \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Experiments are conducted on DBbook2014, comparing LLM-GRec with LightGCN and CIKGRc. The results are shown in Fig. 7. From the overall trend, as ρ increases (i.e., the data becomes

denser), all models exhibit consistent performance improvements in terms of Recall@50 and NDCG@100. This indicates that richer interaction data generally benefits recommendation performance. Under different sparsity levels, LLM-GRec consistently outperforms both LightGCN and CIKGRc across all settings. Notably, in highly sparse scenarios (e.g., $\rho = 0.2$ and $\rho = 0.4$), LLM-GRec maintains a clear performance advantage. For instance, at $\rho = 0.2$, LLM-GRec achieves significant improvements over LightGCN on both Recall@50 and NDCG@100. As the data becomes denser, the performance gap between models gradually narrows; however, LLM-GRec still achieves the best overall performance. These results demonstrate that the semantic interest modeling and multi-granularity representation introduced in LLM-GRec effectively alleviate the limitations caused by sparse interaction data, enabling more stable user preference modeling in low-data regimes. Furthermore, the proposed denoising mechanism and contrastive learning strategy further enhance robustness under incomplete data conditions. Overall, the above findings clearly answer RQ7, confirming that LLM-GRec exhibits superior performance particularly in sparse scenarios.

5. Conclusion

In this paper, we address the challenges of interaction sparsity and insufficient semantic information in recommendation systems by proposing a novel LLM-enhanced

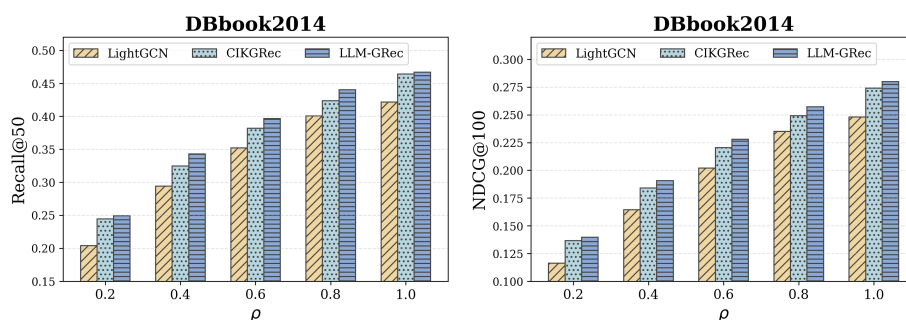


Figure 7: Performance of various models under different sparse interaction ratios.

recommendation framework, namely LLM-GRec. The proposed method leverages LLM to generate semantic user interests, and integrates multi-granularity interest modeling with graph-based learning to improve the expressiveness of user preference representations. In addition, a denoising mechanism and contrastive learning strategy are introduced to enhance robustness under noisy conditions. Specifically, we first design a collaborative-enhanced interest generation mechanism, which alleviates the limitation of incomplete semantic representation caused by relying solely on textual information. Then, we incorporate granular-ball computing to perform multi-granularity interest modeling, which improves the robustness of interest representations under sparse and noisy environments from a structural perspective. Furthermore, we propose a dynamic mask-based denoising strategy to effectively suppress the accumulation and propagation of semantic noise during graph learning. These components work synergistically, enabling the model to learn more stable and discriminative user representations. Extensive experimental results on multiple real-world datasets demonstrate that LLM-GRec consistently outperforms state-of-the-art methods and achieves more significant improvements in sparse scenarios. These findings highlight the importance of integrating semantic information with structural modeling in recommendation systems.

For future work, we plan to explore more fine-grained semantic modeling strategies and develop more efficient online inference mechanisms to further enhance the effectiveness and practicality of the proposed framework.

Acknowledgments:

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- [illegible]

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